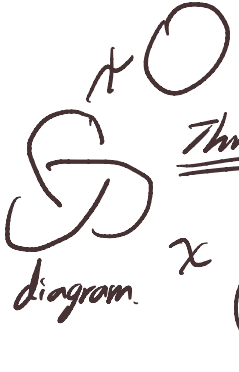
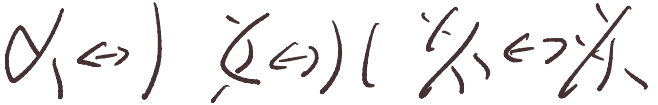
Kauffman bracket and the colored Jones polynomial.

Review

- A knot $K: S^1 \hookrightarrow \mathbb{R}^3$ (or S^3): a smooth embedding up to ambient isotopy.
  link: $S^1 \sqcup S^1 \sqcup \cdots \sqcup S^1$
  $H: \mathbb{R}^3 \times I \to \mathbb{R}^3$ s.t. $H(\cdot, t)$: homeo. on $\mathbb{R}^3$
  - $H(\cdot, 0) = id.$, $H(\cdot, 1)(K_1) = K_2$.

- 
  diagram

  Thm Two links are equiv. iff. corresponding diag.s are equiv. modulo

  Reidemeister moves

- Invariants $\Phi: \{\text{links}\}/_{\sim \text{ambient isotopy}} \to \mathcal{C}$

  $\{\text{diagrams}\}/_{\sim RM.s}$ $\xrightarrow{\tilde{\Phi}}$ $\mathcal{C}$

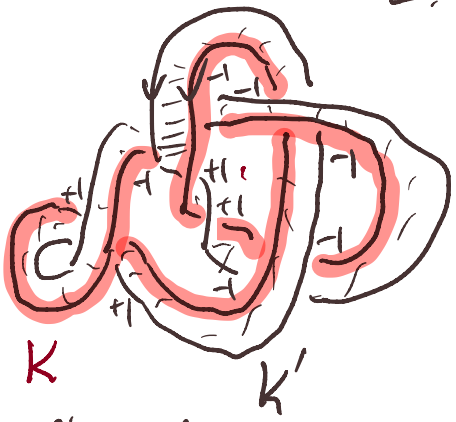
  Ex. Quandle coloring invariant $\mathcal{C} = \mathbb{N} \cup \{\infty\}$

  - Knot quandle $\mathcal{C} = \{\text{groups}\}/_{\text{isom.}}$

Frame Link

Def A framed knot: $\hookrightarrow \mathbb{R}^3$: a smooth embedding up to ambient isotopy.

Link L: $\bigcup \bigcup \cdots \bigcup$



$lk(K, K') = \frac{1}{2}(6\cdot(-1) + 4\cdot(-1)) = -1$

Def (Linking number) $L = K_1 \sqcup K_2$: oriented link.



crossing sign of c: $\mathrm{sgn}(c)$

$$lk(K_1, K_2) = \frac{1}{2}\sum \mathrm{sgn}(c)$$

c: crossing between different components

Linking number.

Exa

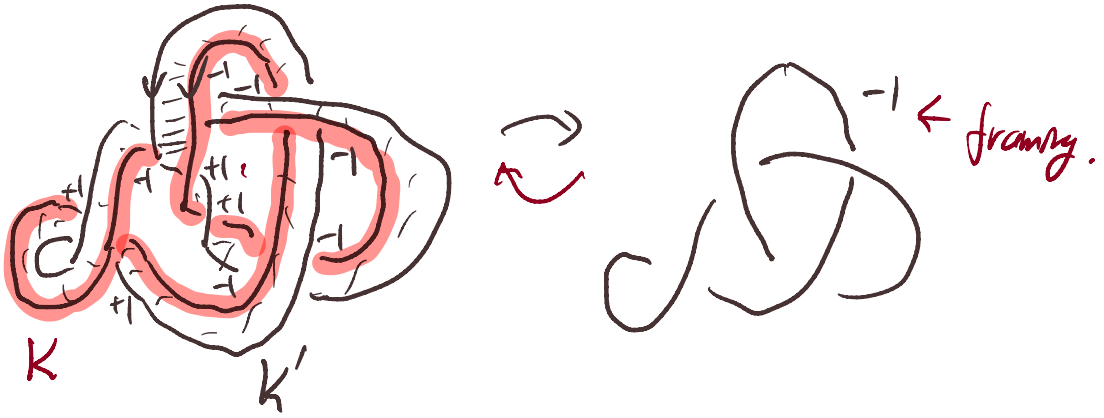
L





$$lk(K_1, K_2) = \frac{1}{2}\sum \mathrm{sgn}(c) = \frac{1}{2}(+1+1) = 1$$

c: crossing between diff. comp



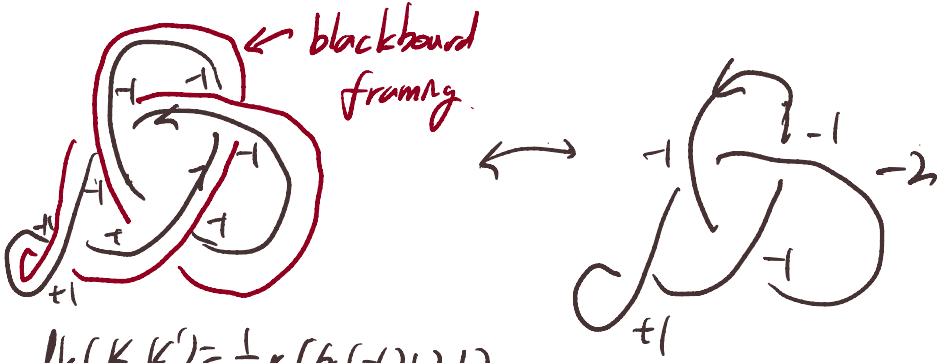
{framed links} $\overset{1-1}{\longleftrightarrow}$ {links w/ integer numbers}

Remark



How to represent framed links by diagrams and which local moves are required?

Blackboard framing



$lk(K, K') = \frac{1}{2} \cdot (6 \cdot (-1) + 2 \cdot 1)$

$= -2$

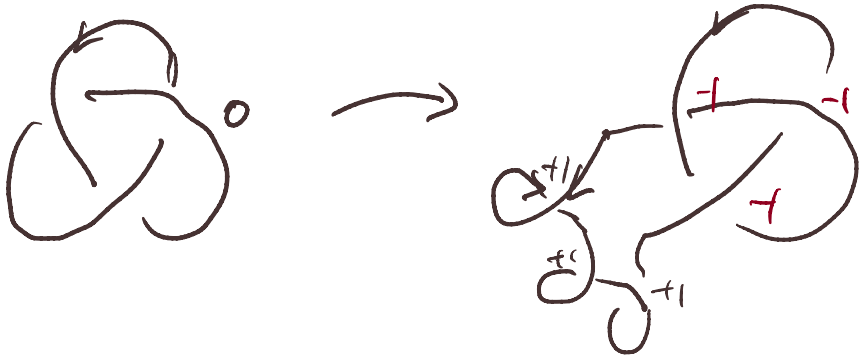
Remark



$lk(K, K') = w(D)$: the writhe of D.

$= \sum_{c : \text{crossing of } D} \text{sgn}(c)$

From the above observation,

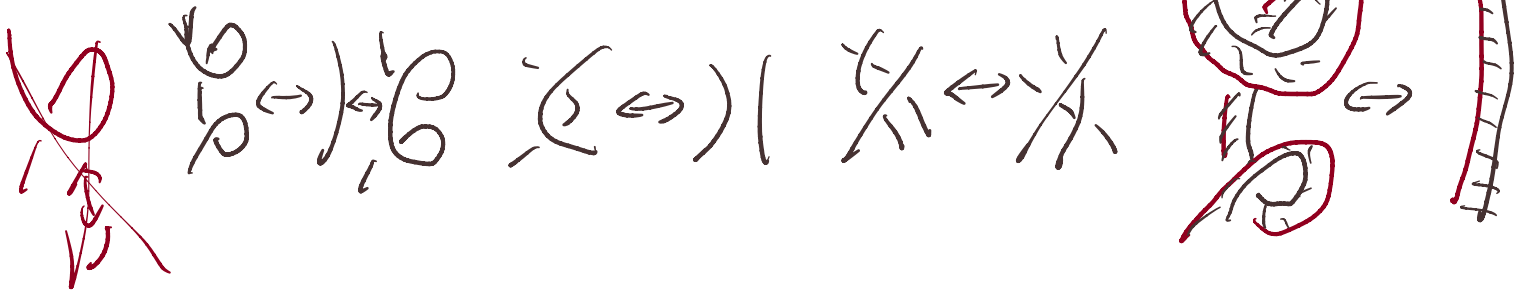


In this case, blackboard framing gives Linking number -3.

A diagram corresponding to a framed knot.

Prop Two framed Links L_1, L_2 are equiv.

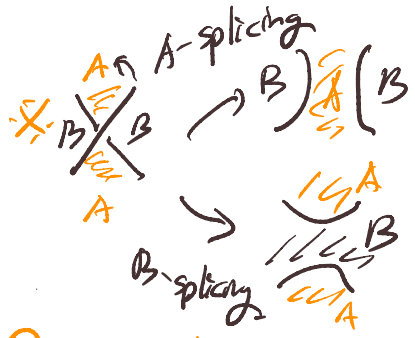
iff. D_{L_2} can be obtained from D_{L_1} by fin. seq. of



Kauffman bracket $\langle\ \rangle' = \frac{\langle\ \rangle}{(-A^2 - A^{-2})}$ $\langle O \rangle' = 1$

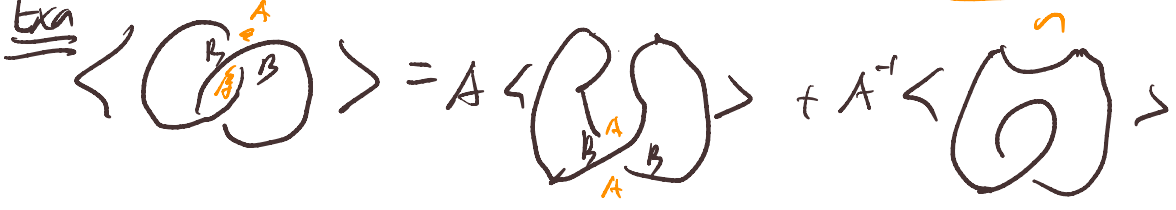
$\langle\ \rangle$: {diagrams} $\to \mathbb{Z}[A^{\pm 1}]$ defined by

- $\langle \times \rangle = A\langle)(\rangle + A^{-1}\langle \asymp \rangle$



- $\langle L \sqcup \bigcirc \rangle = (-A^2 - A^{-2})\langle L \rangle \Rightarrow \langle \underbrace{O, O, \cdots, O}_{n} \rangle = (-A^2 - A^{-2})^{n-1}$.

Exa



$$= A\left(A\langle \text{⌀} \rangle + A^{-1}\langle \text{∞} \rangle\right) + A^{-1}\left(A\langle \text{∞} \rangle + A^{-1}\langle O \rangle\right)$$

$\langle \text{⌀} \rangle = (-A^2 - A^{-2})^2$, $\langle \text{∞} \rangle = (-A^2 - A^{-2})^1$, $\langle \text{∞} \rangle = (-A^2 - A^{-2})^1$, $\langle O \rangle = (-A^2 - A^{-2})^2$

Thm $\langle D \rangle$ is invariant under $R\mathrm{II}$, $R\mathrm{III}$, but not $R\mathrm{I}$

(:) $\langle$  $\rangle = (-A^3)^{\pm 1}\langle D \rangle$ ← w.r.t the sign of kink.

Coro $\langle D \rangle$ is invariant for framed links

Def For an oriented knot diag. D

$$w(D) = \sum_c \mathrm{sgn}(c) : \text{the writhe of } D.$$

Thm $(-A)^{-3w(D)}\langle D \rangle$: invariant under RMIs :

!!

$J(K, A)$ < the Jones polynomial.

Rem $V_0(t) = J(K, t^{-\frac{1}{4}})$: the Jones polynomial.

Generalizations of Jones polynomial

- 2-variable Homflypt, Kauffman 2-variable

$$a\langle \; \rangle - a^{-1}\langle \; \rangle = z\langle \; \rangle \qquad a\langle \; \rangle - a\langle \; \rangle = z\langle \; \rangle$$

Not correct!!!

- Representation theory + quantum group $U_q(sl_2)$

  ⇒ Reshetikhin-Turaev, colored Jones,

  ⋮

Colored Jones polynomial

Goal: To generalize Jones polynomial by modifying diagrams.

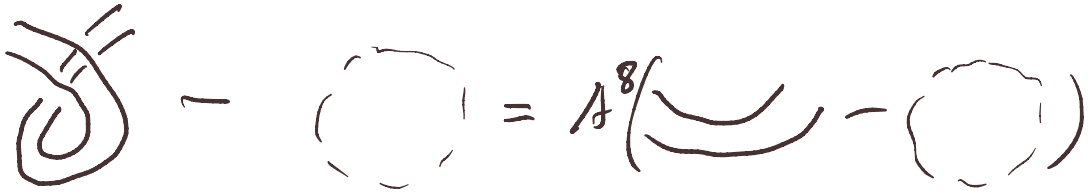


$D^{(1)}$ $D^{(2)}$ $D^{(3)}$

$\langle \bigcirc_1 \rangle = (-A^3)^0 \langle) (\rangle$

$\Rightarrow \langle D^{(n)} \rangle$ is invariant under RM2, 3, but not RM1.





Take $D^{S_2} = D^{(2)} - 1\emptyset \Rightarrow (D^{S_2} \cup O) = (A^8)^{\circ} D^{S_2}$

Thm $(A^8)^{-\mathrm{writhe}(D)} \langle D^{S_2} \rangle$: an invariant under RM.S

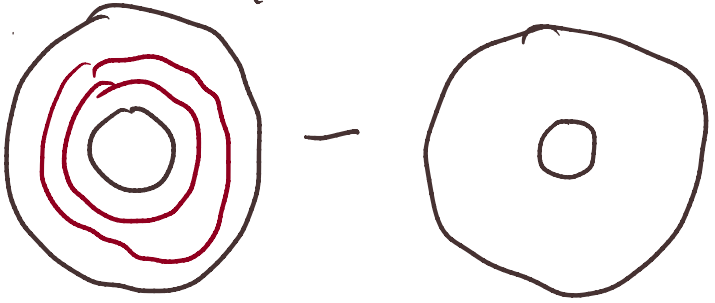
!!

$J_2(K, A)$: 2nd colored Jones polynomial.

For $D^{(k)}$, $k \geq 3$, what is D^{S_k} to obtain a knot invariant?



$D^{(1)}$ — $1 \cdot \phi$ $=: D^{S_2}$



$= S^2$

Such a linear combination of links on annulus provides a lin. comb. of links by putting them on the band for the given diagram.

&

S_2 is "eigenvector" of the operator 

$\Rightarrow$ Let us consider a kind of vector space consisting of links on 

Kauffman bracket skein module

Def M: an oriented 3-mf. (w/ boundary)

R: a comm. ring w/ identity (usually $\mathbb{Z}[A^{\pm 1}]$, $\mathbb{Q}$, $\mathbb{C}$)

$\mathcal{L}^{fr}$: the set of ambient isotopy classes of framed links in M

$R\mathcal{L}^{fr} := $ a free module generated by $\mathcal{L}^{fr} \sqcup \{\emptyset\}$ ($\emptyset$: empty link)

S: a submodule of $R\mathcal{L}^{fr}$ gen. by

$L_+ - A\,L_0 - A^{-2}\,L_\infty$, $\bigcirc - (-A^2 - A^{-2})\,\emptyset$

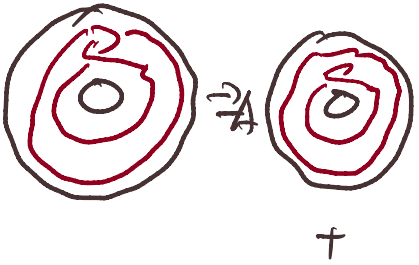
$\Rightarrow S_{2,\infty}(M;R) := R\mathcal{L}^{fr}/S$: called the Kauffman bracket skein module.

$S_{2,\infty}(\Sigma \times [0,1])$

Kauffman bracket skein algebra of $\Sigma \times I$, $S^A(\Sigma)$

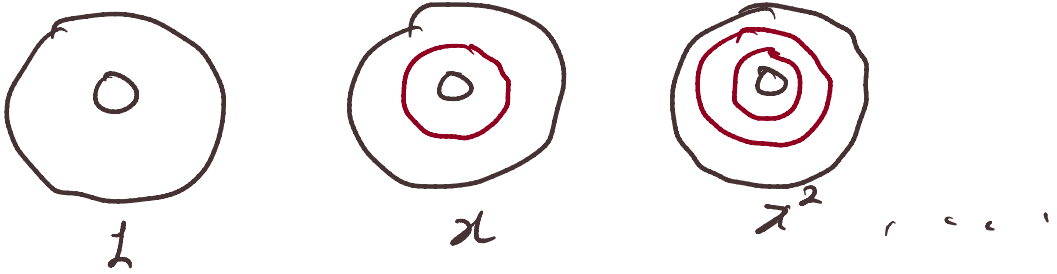
is $S_{2,\infty}(\Sigma \times I)$ w/ multiplication by stacking links.





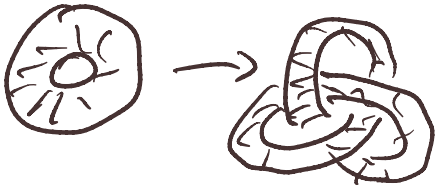
Thm $S^A(\Sigma)$: inf. dim. module spanned by $\{1, x, x^2, \dots\}$

fin. gen. by x, $S^A(\Sigma_1) \cong \mathbb{Z}[A^{\pm 1}][x]$.



Threading operation

D: a diag. of K, $p(x) \in S^A(\bigcirc)$

$\Rightarrow D^{p(x)}$ is induced by i: 

If $p(x)=x^n$, D^{x^n}: a n-parallel of D.

If $p(x)=\sum c_i x^i$, $D^{p(x)}=\sum_{i=1}^{n} c_i D^{x^i}$.

Remark $J_2(K,A)=(A^8)^{-writhe}\langle D^{x^2-1}\rangle$

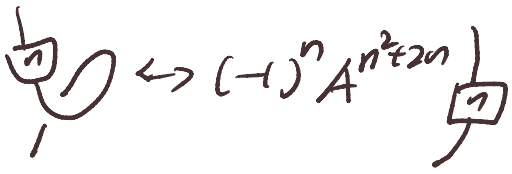
Why D^{x^2-1} worked? 

i.e. x^2-1 is an "eigenvector" for RMI.

Chebyshev polynomial of 2nd type

$S_0(x)=1,\ S_1(x)=x,\ S_{n+1}(x)=xS_n(x)-S_{n-1}(x)$ in $S^3(\textcircled{F})$

Denote it by  $=: S_n(x)$

Lemma  $\leftrightarrow (-1)^n A^{n^2+2n}$

Coro $J_n(K,A) = ((-1)^n A^{n^2+2n})^{-wr(D)} \langle D^{S_n} \rangle$

: invariant under RMs. called n-th colored Jones polynomial.

Exa $J_n(O;A) =$  $= (-1)^n \dfrac{A^{2n+2}-A^{-2n-2}}{A^2-A^{-2}} =: \Delta_n$

Remark • If you are familiar w/ Temperley-Lieb algebra,

then 

$\cong S_n$ is the closure of Jones-Wenzl idempotent.

- The 1st Jones polynomial is original Jones polynomial.
- The colored Jones polynomial is known as quantum invariant.